

# Wei Wang (王尉)

Ph.D. Candidate in Mathematics, School of Mathematical Sciences, Peking University

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## EDUCATION

### Peking University, School of Mathematical Sciences

Beijing, China

Ph.D. Candidate in Mathematics

Sept. 2022 – 2027 (expected)

Advisor: Prof. Zhifei Zhang

### Zhejiang University, School of Mathematical Sciences

Hangzhou, Zhejiang, China

B.S. in Mathematics and Applied Mathematics

Sept. 2018 – June 2022

Advisor: Prof. Ting Zhang

Thesis: *Homogenization Problem of Second Order Elliptic Systems with Lower Order Terms*

## RESEARCH INTERESTS

Geometric variational problems; elliptic and parabolic partial differential equations; harmonic maps and their flows; liquid crystal theory; homogenization theory; nonlocal operators.

## PUBLICATIONS AND ACCEPTED PAPERS

- (7) Landau–de Gennes model with sextic potentials: asymptotic behavior of minimizers.  
**Wei Wang** and Zhifei Zhang.  
Accepted for publication in *Memoirs of the European Mathematical Society*. [[arXiv: 2404.00677](#)]
- (6) Existence of weak solutions for two-phase matrix-valued harmonic map flows.  
**Wei Wang**, Wei Wang (王伟), and Zhifei Zhang.  
*Journal of Functional Analysis* 291 (2026), no. 11, Article No. 111623. [[doi](#)] [[arXiv: 2506.16278](#)]
- (5) Uniform estimates of Landau–de Gennes minimizers in the vanishing elasticity limit with line defects.  
Haotong Fu, Huaijie Wang, and **Wei Wang**.  
*SIAM Journal on Mathematical Analysis* 58 (2026), no. 4, 3306–3324. [[doi](#)] [[arXiv: 2508.01811](#)]
- (4) Stratification and rectifiability of harmonic map flows via tangent measures.  
Haotong Fu, **Wei Wang**, Ke Wu, and Zhifei Zhang.  
*Mathematische Annalen* 395 (2026), Article No. 92. [[doi](#)] [[arXiv: 2504.14880](#)]
- (3) On the sharp constants in the regional fractional Sobolev inequalities.  
Rupert L. Frank, Tianling Jin, and **Wei Wang**.  
*SN Partial Differential Equations and Applications* 6 (2025), Article No. 15. [[doi](#)] [[arXiv: 2403.00357](#)]
- (2) Uniform estimates of resolvents in homogenization theory of elliptic systems.  
**Wei Wang**.  
*Journal of Differential Equations* 370 (2023), 1–65. [[doi](#)] [[arXiv: 2211.14614](#)]
- (1) Homogenization theory of elliptic systems with lower order terms in dimension two.  
**Wei Wang** and Ting Zhang.  
*Communications on Pure and Applied Analysis* 22 (2023), no. 3, 787–824. [[doi](#)] [[arXiv: 2210.06329](#)]

- (14) Asymptotics for two-dimensional parabolic vectorial Allen–Cahn systems.  
Zhiyuan Dai, Haotong Fu, Huaijie Wang, and **Wei Wang**.  
Preprint. [[arXiv: 2609.27596](#)]
- (13) Small-energy regularity for stationary weak solutions of the Liouville equation.  
Haotong Fu, Kelei Wang, **Wei Wang**, and Ke Wu.  
Preprint. [[arXiv: 2609.26026](#)]
- (12) Boundary asymptotics for two-dimensional vectorial Allen–Cahn systems.  
Zhiyuan Dai, Haotong Fu, Huaijie Wang, and **Wei Wang**.  
Preprint. [[arXiv: 2609.00739](#)]
- (11) Boundary regularity for Oseen–Frank minimizers with arbitrary positive splay, twist, and bend constants.  
Zhiyuan Dai, Haotong Fu, Huaijie Wang, and **Wei Wang**.  
Preprint. [[arXiv: 2608.22287](#)]
- (10) Sharp instability of planar isotropic–nematic interfaces in the Landau–de Gennes model.  
**Wei Wang** and Qin Wu.  
Preprint. [[arXiv: 2608.12131](#)]
- (9) Smooth convergence of Landau–de Gennes minimizers in the Lyuksyutov regime.  
Haotong Fu, Huaijie Wang, and **Wei Wang**.  
Preprint. [[arXiv: 2608.08456](#)]
- (8) On the absence of point defects in biaxial Landau–de Gennes models.  
Haotong Fu, Huaijie Wang, **Wei Wang**, and Zhifei Zhang.  
Preprint. [[arXiv: 2608.04908](#)]
- (7) Singular limits for three-dimensional global minimizers of Ginzburg–Landau-type functionals: uniform estimates and singular sets.  
Giacomo Canevari, Haotong Fu, and **Wei Wang**.  
Preprint. [[arXiv: 2606.27691](#)]
- (6) A note on an approximate fixed point theorem for VMO maps.  
Haotong Fu and **Wei Wang**.  
Preprint. [[arXiv: 2606.15670](#)]
- (5) Sharp convergence rates for parabolic Green’s functions in time-independent periodic homogenization.  
**Wei Wang**.  
Preprint. [[arXiv: 2606.12101](#)]
- (4) Asymptotics of minimizers for Ginzburg–Landau-type functionals in high dimensions.  
Giacomo Canevari, Haotong Fu, and **Wei Wang**.  
Preprint. [[arXiv: 2605.04442](#)]
- (3) Improved convergence of Landau–de Gennes minimizers in the vanishing elasticity limit.  
Haotong Fu, Huaijie Wang, and **Wei Wang**.  
Preprint. [[arXiv: 2507.14955](#)]
- (2) Fine structure of rupture set for semilinear elliptic equations with singular nonlinearity.  
**Wei Wang** and Zhifei Zhang.  
Preprint. [[arXiv: 2411.16048](#)]

- (1) Quantitative stratification and sharp regularity estimates for supercritical semilinear elliptic equations.

Haotong Fu, **Wei Wang**, and Zhifei Zhang.

Preprint. [[arXiv: 2408.06726](https://arxiv.org/abs/2408.06726)]

#### RESEARCH VISITS AND PROGRAMS

##### **University of Verona, Department of Computer Science**

*Verona, Italy*

*Visiting Ph.D. Student*

Mar. – Apr. 2026

Host: Prof. Giacomo Canevari

##### **HKUST, Department of Mathematics**

*Hong Kong, China*

*Summer Research Program*

June – Sept. 2021

Advisor: Prof. Tianling Jin

#### SELECTED TALKS

- Singular Limits for Three-Dimensional Global Minimizers of Ginzburg–Landau-Type Functionals: Uniform Estimates and Singular Sets  
Conference on Geometric Measure Theory and Partial Differential Equations  
*Enshi, Hubei, China, Aug. 2026.*
- Uniform Estimates for Landau–de Gennes Minimizers with Point and Line Defects  
Academic Forum on Theory and Numerical Analysis of Liquid Crystals and Complex Systems  
*Xiangtan University, July 2026.*
- Landau–de Gennes Model with Sextic Potentials: Asymptotic Behavior of Minimizers  
Academic Workshop on Fluid Dynamics Equations  
*Peking University, Jan. 2026.*
- Stratification for Harmonic Map Flows via Tangent Measures: Rectifiability and Regularity  
Jinhua Conference on Harmonic Analysis and Its Applications  
*Zhejiang Normal University, Nov. 2025.*
- Improved Convergence of Landau–de Gennes Minimizers in the Vanishing Elasticity Limit  
Conference on Theory and Computation of Quasi Crystals  
*Nankai University, Aug. 2025.*
- Rectifiability of the Stratification for Harmonic Map Flows via Tangent Measures  
PDE and Geometry Seminar  
*Zhejiang University, June 2025.*

#### GRANTS AND PROJECTS

- **Physical Modelling and Mathematical Theory of Liquid Crystals** 2023 – 2028  
Participant, NSFC, Grant No. 12288101.
- **Physical Modeling and Mathematical Theory of Liquid Crystal Systems** 2023 – 2028  
Participant, National Key R&D Program of China, Grant No. 2023YFA1008801.

#### ACADEMIC SERVICE

##### **Mathematical Reviews / MathSciNet, American Mathematical Society**

Registered prospective reviewer.

## TEACHING EXPERIENCE

- **Peking University, Teaching Assistant** 2022 – 2026  
Advanced Mathematics (A, B, and C); Real Analysis; Functional Analysis (I); Harmonic Analysis.
- **Zhejiang University, Teaching Assistant** 2021 – 2022  
Mathematical Analysis (I and II).

## HONORS AND AWARDS

- First Prize, Graduate Research Exhibition Apr. 2025  
*School of Mathematical Sciences, Peking University*
- Third Prize, Graduate Research Exhibition Apr. 2024  
*School of Mathematical Sciences, Peking University*
- Elite Program for Graduate Students in Mathematics Oct. 2023  
*Peking University*
- Nomination Award, ICCM Creative Undergraduate Thesis Aug. 2022  
*Shing-Tung Yau Center of Southeast University*
- First Prize, 11th Chinese Mathematics Competitions Finals Apr. 2021  
*Wuhan University*
- Nomination Award, S.-T. Yau College Student Mathematics Contest June 2021  
*Yau Mathematical Sciences Center, Tsinghua University*

**Skills:** LaTeX, MATLAB, Mathematica, Python, Lean.    **Languages:** Chinese (native), English (fluent).